

Assignment 6

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First, we install the relevant packages.

```
library(tidyverse)
library(here)
library(devtools)
library(testthat)
```

Next, we call in the functions from our repository.

```
#Call functions

source("climate_conditions.R")
source("land_conditions.R")
source("wildfire_likelihood.R")
```

Run Functions with Example Data

Here we generate sample site conditions to run functions on and then proceed to run each function accordingly. Function results are each stored in a new corresponding vector.

```
# Example site conditions
temperature <- 35
humidity <- 15
wind_speed <- 40
fuel_moisture <- 6
slope <- 25

# Run each function once
```

```

climate_result <- climate_conditions(
  temperature = temperature,
  humidity = humidity,
  wind_speed = wind_speed
)

land_result <- land_conditions(
  fuel_moisture = fuel_moisture,
  slope = slope
)

wildfire_result <- wildfire_likelihood(
  climate_fire_probability = climate_result,
  land_fire_probability = land_result
)

# View function results
climate_result

```

```
[1] 0.7972222
```

```
land_result
```

```
[1] 0.002405574
```

```
wildfire_result
```

```
[1] 0.001917777
```

Run tests

Finally, we run all of our new tests which are stored in the `tests` folder of our repository.

```
test_dir(here("tests"))
```

```
v | F W S OK | Context
```

```
/ |          0 | climate_conditions
```

```

v |          5 | climate_conditions

/ |          0 | land_conditions
x | 4          5 | land_conditions
-----
Failure ('test_land_conditions.R:18:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = -5, slope = 1)` to throw a error.

Failure ('test_land_conditions.R:21:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 101, slope = 1)` to throw a error.

Failure ('test_land_conditions.R:27:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 10, slope = 100)` to throw a error.

Failure ('test_land_conditions.R:30:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 10, slope = -100)` to throw a error.
-----

/ |          0 | wildfire_likelihood
x | 5          4 | wildfire_likelihood
-----
Failure ('test_wildfire_likelihood.R:16:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 1.01, land_fire_probability = 0.2)`

Failure ('test_wildfire_likelihood.R:20:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = -1, land_fire_probability = 0.2)` to

Failure ('test_wildfire_likelihood.R:28:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 0.2, land_fire_probability = 1.01)`

Failure ('test_wildfire_likelihood.R:32:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 0.2, land_fire_probability = -
1)` to throw a error.

Failure ('test_wildfire_likelihood.R:58:3'): wildfire_likelihood_works
Expected `... < 1` to be TRUE.
Differences:
`actual`:  FALSE
`expected`: TRUE
-----

== Results ==

```

```

-- Failed tests -----
Failure ('test_land_conditions.R:18:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = -5, slope = 1)` to throw an error.

Failure ('test_land_conditions.R:21:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 101, slope = 1)` to throw an error.

Failure ('test_land_conditions.R:27:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 10, slope = 100)` to throw an error.

Failure ('test_land_conditions.R:30:3'): land_conditions_checks_errors
Expected `land_conditions(fuel_moisture = 10, slope = -100)` to throw an error.

Failure ('test_wildfire_likelihood.R:16:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 1.01, land_fire_probability = 0.2)` to throw an error.

Failure ('test_wildfire_likelihood.R:20:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = -1, land_fire_probability = 0.2)` to throw an error.

Failure ('test_wildfire_likelihood.R:28:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 0.2, land_fire_probability = 1.01)` to throw an error.

Failure ('test_wildfire_likelihood.R:32:3'): wildfire_likelihood_checks_errors
Expected `wildfire_likelihood(climate_fire_probability = 0.2, land_fire_probability = -1)` to throw an error.

Failure ('test_wildfire_likelihood.R:58:3'): wildfire_likelihood_works
Expected `... < 1` to be TRUE.
Differences:
`actual`:  FALSE
`expected`: TRUE

[ FAIL 9 | WARN 0 | SKIP 0 | PASS 14 ]

Error:
! Test failures.

```

Test results

1. `test_climate_conditions.R`

This file runs the following tests to: check that the function returns a single numeric probability-like value; check that the value returned is between 0 and 1; and confirm that the function behaves logically by producing a higher climate fire probability under hotter, drier, and windier conditions.

These tests are valid and should pass.

2. `test_land_conditions.R`

The land conditions tests check both input handling and expected fire-risk behavior. The `expect_error()` tests intentionally pass unrealistic inputs to see whether the function breaks appropriately. The remaining tests check that lower fuel moisture and steeper slopes increase land-based fire probability, which matches the assumptions of the model.

3. `test_wildfire_likelihood.R`

The wildfire likelihood tests check whether the function handles invalid probability inputs and whether the calculation behaves as expected. The error tests use values outside the 0–1 probability range or non-numeric inputs to demonstrate how the function responds to bad inputs. The remaining tests confirm that increasing either climate fire probability or land fire probability increases the final wildfire probability.